(a) A JFET amplifier has been biased as shown in Fig. 2.1. As a design engineer, determine the appropriate value of the drain resistance (R_D) if the amplifier needs an AC gain of 10.

[15 marks]

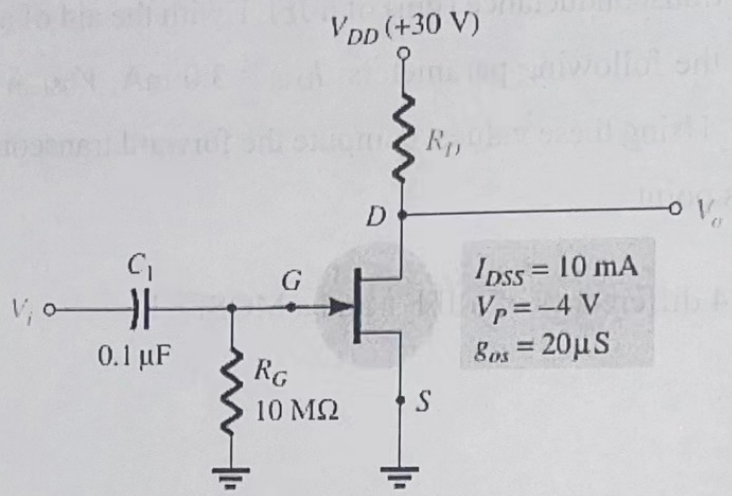


Fig 2.1: JFET amplifier

(b) A design engineer designed a 2-stage amplifier as shown in Fig 2.2. The engineer adopted V_{DD}=10V, the drain resistance R_D=10kΩ, the gate resistance R_G =1MΩ, the source resistance R_S= 1kΩ for the design. If the transconductance value for both JFET is set to 1 mS, then as a microelectronic engineer, derive the expression for the overall gain of this circuit (using DC and AC analysis) and find the value of the output voltage V_0 if you connect a 0.5V (p-p) ac source at the input terminal. Assume that the capacitors are shorts at the frequency of operation.

[10+5=15 marks]

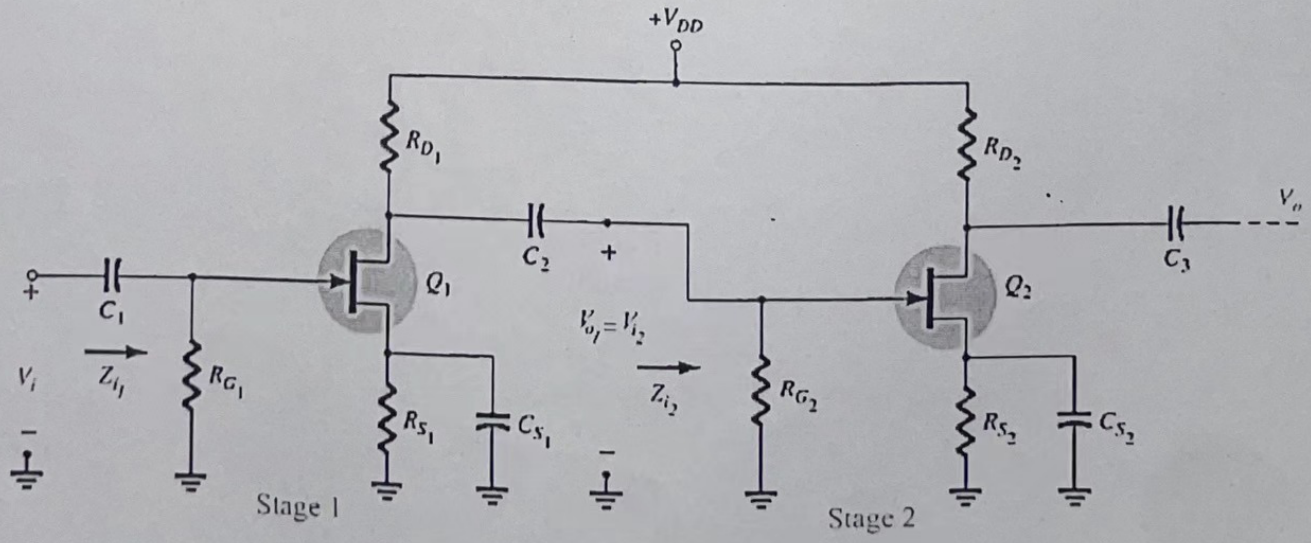


Fig. 2.2: A two stage JFET amplifier

Question 3 (30 marks) [CLO 3, PLO 3, C5]

(a) Given a two stage operational amplifier circuit in Fig 3.1, evaluate the values of v_1, v_2, v_o, i_s, and i_r in terms of the source voltage (v_s). Consider that the source resistance Rs= 1kΩ and the feedback resistance R_f is 10 kΩ. **[5x2=10 marks]**

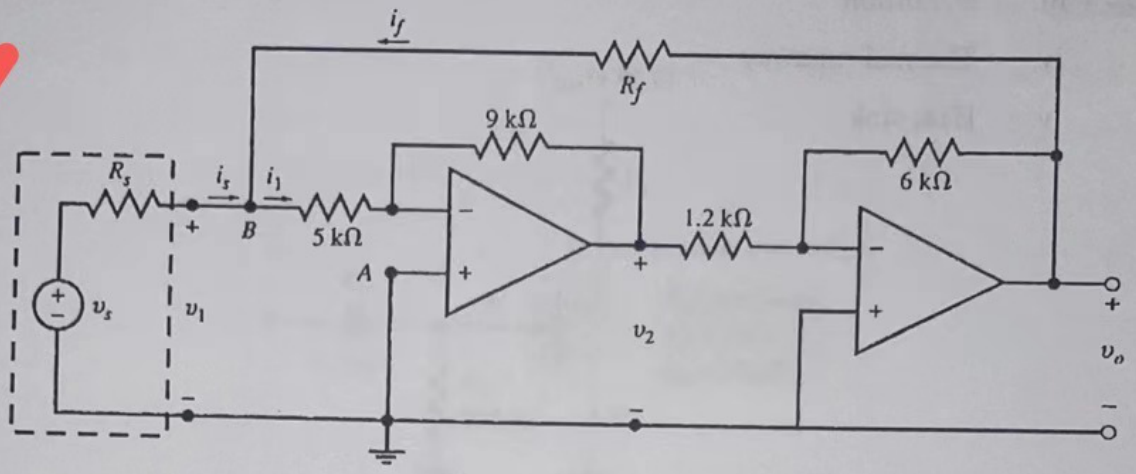


Fig. 3.1 A two stage opamp circuit

(b) Estimate the values of resistances (R_1 to R_4) used in an opamp based subtractor circuit (need to draw the circuit and using circuit analysis, derive the equation of output voltage) that will provide a zero output voltage if both inputs are of same magnitude. **[10 marks]**

(c) Evaluate the cut-off frequencies of the opamp based bandpass filter shown in Fig 3.2. Besides, estimate the output voltage (V_0) if a 1 mV AC signal is applied at the input terminal (V_i). Assume that R_1=R_2=R_G=R_F=1kΩ, C1=1μF and C2=0.02 μF.

[3+3+4=10 marks]

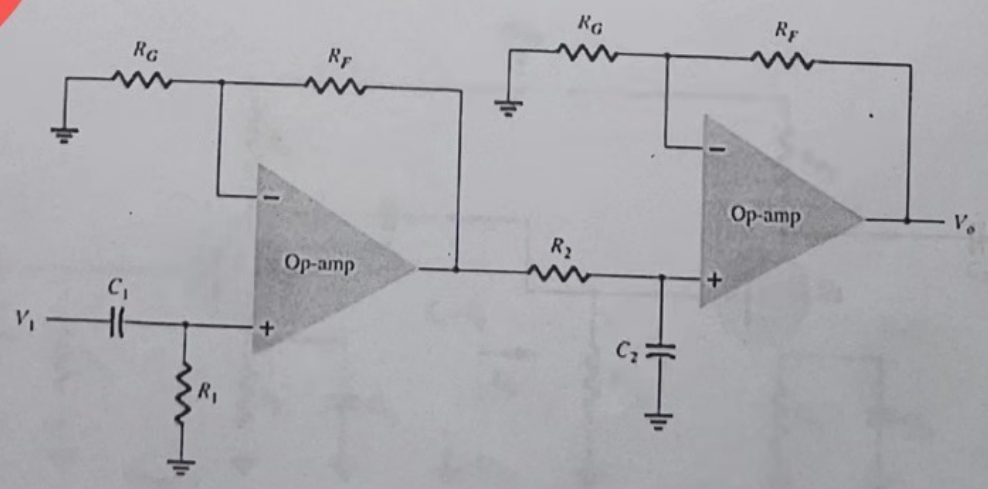


Fig. 3.2 A bandpass filter circuit